

# Notes: Milgrom (JPE, 1988)

## Employment Contracts, Influence Activities, and Efficient Organization Design

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# 1 Big picture

## 1.1 One-sentence summary

This note summarizes Milgrom (JPE, 1988), which develops a theory of **influence costs** inside organizations: because efficient employment contracts usually do not fully compensate workers for post-hiring decisions and job characteristics, employees care about managerial discretion and may waste productive time trying to influence decision makers; efficient organization design therefore restricts discretion especially when a decision has large distributional consequences for employees but little direct importance for firm value.

## 1.2 Central research question

The paper asks why centralized authority is costly even when a central decision maker could, in principle, make better informed decisions. The answer is not simply that central managers make mistakes. The deeper mechanism is that central authority creates a prize over which affected agents compete. When workers care about the allocation of jobs, assignments, projects, promotions, offices, routes, or other post-hiring decisions, they spend time and resources trying to shape the decision. These efforts are privately rational but socially wasteful when they are mainly redistributive.

## 1.3 Core economic mechanism

The logic has three steps:

1. **Employment contracts are incomplete in a substantive sense.** Even optimal contracts do not generally pay full compensating differentials for every possible post-hiring event or job attribute.
2. **Employees therefore care about managerial decisions.** They prefer good working conditions, human-capital-building assignments, or “critical” jobs that pay more because quits are costly to the employer.
3. **Managerial discretion then invites influence activities.** If managers can make decisions that redistribute utility across employees, employees may lobby, persuade, delay, withhold information, or otherwise spend time influencing outcomes.

## 1.4 Main organizational-design implication

An efficient organization should not always give managers full discretion. It should restrict discretion, use rules, or precommit to procedures when the potential informational or coordination benefit of discretion is small relative to the redistributive stakes created by the decision.

# 2 Incremental contribution of the paper

## 2.1 Contribution relative to the theory of the firm

The paper adds an internal political-economy channel to the theory of the firm. Coase (1937) asked why production is not conducted inside one large firm, and transaction-cost theories emphasize opportunism, ownership, and asset specificity. Milgrom adds another cost of centralization: broader authority creates more opportunities for agents to seek redistributive rents inside the organization.

## 2.2 Contribution relative to rent-seeking theory

Earlier rent-seeking models focused mainly on public policy: tariffs, monopolies, subsidies, and regulation. This paper transfers the same logic into private organizations. Influence costs can arise not only in legislatures or agencies but also in firms, courts, acquisition contests, procurement decisions, and internal job assignments.

## 2.3 Contribution relative to employment-contract theory

The paper does not assume, without explanation, that workers care about decisions. It derives this property from several contracting models. Even when contracts are efficient, workers may not be fully compensated for post-hiring job conditions, human-capital effects, or turnover-sensitive wage premia. This makes internal influence a natural consequence of optimal contracting rather than merely a behavioral anomaly.

## 2.4 Contribution relative to organization design

The paper provides a clear design principle: discretion is most valuable when the central decision can improve firm profits substantially, and most dangerous when it redistributes large rents among employees. This explains why firms use seniority rules, standard operating procedures, bright-line rules, and other discretion-limiting devices.

## 2.5 Table 1: Incremental contribution and why it matters

| Dimension               | What the paper adds                                                                                   | Why it matters                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Theory of the firm      | Centralization creates influence costs, not only coordination benefits.                               | Explains why firms and economies do not simply centralize all decisions.                                             |
| Rent seeking            | Rent seeking is generalized from public policy to internal firm decisions.                            | Makes lobbying-like behavior a core organizational problem.                                                          |
| Employment contracts    | Efficient contracts often fail to fully compensate post-hiring job characteristics.                   | Provides micro-foundations for why employees care about assignments and managerial discretion.                       |
| Organization design     | Discretion should be restricted when redistributive stakes are high and firm-level importance is low. | Explains seniority rules, standard operating procedures, limited managerial discretion, and bright-line legal rules. |
| Institutional economics | Institutions should channel self-interest away from redistribution and toward productive activity.    | Links firm design, bureaucratic rules, litigation, innovation, and economic development.                             |

## 3 Why employees care about post-hiring decisions

### 3.1 The contract problem

The paper begins with a Coase-Simon style employment problem. At the contracting date, neither the firm nor the worker knows the exact job conditions that will later be realized. Let the possible

post-hiring states be  $x_1, \dots, x_N$  with probabilities  $p_1, \dots, p_N$ . The contract may specify a wage  $w_i$  for each state. The worker has utility  $u_i(w)$  in state  $i$ , and the firm receives gross revenue  $\pi_i$ .

The efficient contract solves:

$$\begin{aligned} \max_{\{w_i\}_{i=1}^N} \quad & \sum_{i=1}^N p_i(\pi_i - w_i) \\ \text{s.t.} \quad & \sum_{i=1}^N p_i u_i(w_i) \geq \bar{u}, \\ & u_j(w_j) \geq \hat{u} \quad \text{for all } j = 1, \dots, N. \end{aligned}$$

The first constraint is the ex ante participation constraint. The second is a no-quitting constraint after the worker observes the assigned job condition.

### 3.2 Theorem 1: full compensating differentials are special, not generic

Theorem 1 asks when an optimal contract makes the employee indifferent among all post-hiring outcomes. The result is restrictive: full compensation occurs only if preferences can be represented by utility functions satisfying a vertical-translation condition of the form

$$u_i(w) = u_1(w + g_i) \quad \text{for all } w.$$

This means that all job conditions differ only by a fixed wage translation. In most preference specifications, this condition fails. Therefore, efficient contracts generally do not make workers indifferent across job assignments.

**Economic intuition.** Optimal contracting equalizes marginal utility of income across states, not total utility across job characteristics. If different job conditions change how workers value consumption or interact with wage risk, the optimal wage schedule does not fully neutralize those differences.

### 3.3 Key observation: worker rankings need not match firm rankings

A central point follows immediately: the optimal wage schedule depends on the worker's utility and participation constraints, not directly on the firm's gross revenue ranking  $\pi_i$ . Hence the worker's preferred post-hiring state need not be the firm's preferred state. This is the first source of conflict that makes influence activities possible.

## 4 Three reasons full compensating differentials fail

### 4.1 Table 2: Three contract models and their influence implications

| Model                   | Key assumption                                                   | Employee preference created by optimal contract                                                                                   | Influence implication                                                                                          |
|-------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Good working conditions | Job attribute $x$ directly enters utility with wage $w$ .        | Workers prefer better working conditions because bad conditions are not fully compensated.                                        | Workers lobby for pleasant assignments, offices, routes, colleagues, tasks, or conditions.                     |
| Human capital           | Some assignments raise future productivity.                      | Workers prefer assignments that build transferable human capital because future wage gains are not offset by lower current wages. | Workers seek training, visible projects, and skill-building roles even if these are not firm-value-maximizing. |
| Critical jobs           | Some jobs are costly for the firm to refill if the worker quits. | Workers prefer critical jobs because efficient wages rise with quit costs.                                                        | Workers seek roles that increase bargaining power, centrality, or replacement difficulty.                      |

### 4.2 Example 1: good working conditions

Suppose  $x \geq 0$  is a working condition or on-the-job consumption, and  $w \geq 0$  is the wage. With Cobb-Douglas ordinal preferences  $x^\alpha w$  and constant relative risk aversion, the optimal wage schedule can take the form

$$w(x) = \lambda x^{\alpha(1-\beta)/\beta}.$$

When  $\beta < 1$ , the wage is increasing in  $x$ , meaning that better working conditions may be paired with higher wages. This sharply contradicts the standard compensating-differential intuition that worse jobs must always pay more.

**Economic message.** Workers can rationally prefer better working conditions even under efficient contracts. The employment contract does not fully neutralize assignment preferences.

### 4.3 Theorem 2: risk aversion pushes contracts toward fuller compensation

Theorem 2 compares two workers with the same ordinal preferences but different risk aversion. As risk aversion rises, the optimal contract raises wages in low-utility assignments and lowers wages in high-utility assignments. In the limit, extreme risk aversion pushes wages toward full compensating differentials.

**Economic message.** Full compensation is not generally optimal, but stronger risk aversion makes wage smoothing and utility smoothing more important. This explains why the degree of compensation depends on the risk environment and worker preferences.

#### 4.4 Example 2: preference to accumulate human capital

The worker has a two-period life. In the first period, one assignment raises second-period productivity from  $p$  to  $q > p$ , while another leaves productivity unchanged. In the second period, the worker can leave for a wage equal to current marginal product. The optimal contract pays the worker current marginal product in each period. Therefore, a human-capital-building assignment raises future wages but is not offset by a current wage reduction.

**Economic message.** Workers prefer assignments that build general human capital. This can conflict with the employer’s ranking of assignments, because the firm may not capture the future wage gains.

#### 4.5 Example 3: preference for critical jobs

The third example is an efficiency-wage model. Some jobs are more “critical” because worker quits impose larger costs  $\Delta_i$  on the firm. The optimal wage satisfies a condition of the form

$$\Delta_i = (w_i - b) + (1 - \lambda) \frac{F(w_i - b)}{f(w_i - b)},$$

so wages increase with the quit cost  $\Delta_i$ .

**Economic message.** Workers prefer critical jobs because they pay more. A firm’s internal allocation of critical assignments can therefore become a contested prize.

### 5 Influence activities and restricted discretion

#### 5.1 Time allocation between production and influence

The model in Section III formalizes influence activities. A worker has total time  $T$ , split between productive effort  $t$  and influence activity  $s$ , with

$$s + t \leq T, \quad s, t \geq 0.$$

Productive effort raises the probability of high output,  $p(t)$ . Influence effort raises the probability that the central decision maker authorizes a change,  $q(s)$ .

#### 5.2 Organizational importance and redistributive stakes

The model has two key parameters:

- $I$ : the **importance** of managerial discretion to the organization. It measures the potential profit improvement from informed discretionary decision-making.
- $k$ : the **redistributive value** of the decision to the employee. It measures how much utility is transferred to the employee if the decision is made in their favor.

The employee values the favorable change by  $k$ , but this benefit is excluded from the social objective because it is assumed to be a transfer from another employee.

#### 5.3 Why wage incentives may not solve the problem

The firm can design wages that depend on output and whether the change is approved. However, this is costly because the same wage instruments must also provide incentives for productive effort. The firm faces a moral-hazard trade-off: stronger incentives against influence activities may distort productive effort or require costly compensation.

#### 5.4 Theorem 4: when discretion should be eliminated

Theorem 4 establishes that there exist values of  $(I, k)$  such that eliminating management discretion is better than allowing discretion and trying to control influence activities through wages. Moreover, if eliminating discretion is optimal at  $(I, k)$ , then it remains optimal when the organizational importance  $I$  falls or the redistributive stake  $k$  rises.

**Economic message.** The design rule is monotone: low-importance and high-redistribution decisions should be governed by rules rather than by discretion.

#### 5.5 Table 3: The organizational-design rule

| Case                | Decision characteristics                                                    | Influence-cost prediction                                          | Efficient design response                                                         |
|---------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| High $I$ , low $k$  | Decision matters for firm value and redistributes little.                   | Influence costs are limited relative to benefits.                  | Delegate discretion to informed managers.                                         |
| High $I$ , high $k$ | Decision matters for firm value but creates large private stakes.           | Influence is costly, but discretion may still be worth preserving. | Use discretion with safeguards, objective criteria, documentation, or incentives. |
| Low $I$ , low $k$   | Decision has small firm-value effect and small redistributive stakes.       | Influence costs are small.                                         | Either simple rules or discretion can work.                                       |
| Low $I$ , high $k$  | Decision has small firm-value effect but large redistributive consequences. | Influence costs can dominate any informational gain.               | Restrict discretion; use rules, seniority, rotation, or fixed procedures.         |

### 6 Applications and economic interpretation

#### 6.1 Flight-attendant route assignments

The paper uses American Airlines' route-assignment system as an example. Flight attendants bid for routes, and conflicts are resolved by seniority. The airline cares relatively little about which attendant serves which route, but flight attendants care a great deal. A discretionary assignment system would invite lobbying. A seniority rule reduces influence costs.

#### 6.2 Innovation and permissionless experimentation

The paper interprets Western economic development partly through limited interference with innovators. When the social cost of a failed experiment is small but the redistributive consequences of a successful innovation are large, giving a central authority discretion to block innovation creates large influence costs. Permissionless innovation can therefore be efficient.

#### 6.3 Resistance to change

Employees resist changes that threaten their rents, working conditions, status, or future opportunities. This resistance is not merely irrational inertia. It is rent protection arising from incomplete compensation for the distributional consequences of change.



## 6.4 Vertical integration

If suppliers have market power, they may spend resources selling, persuading, and influencing the buyer. Vertical integration can restrict the buyer’s discretion about whom to purchase from, thereby reducing influence costs. This gain must be balanced against the costs of centralizing other decisions.

## 6.5 Takeovers and golden parachutes

Takeover contests can include rent-seeking and rent-protection. Golden parachutes can be interpreted as compensation arrangements that reimburse displaced managers for lost rents, thereby reducing inefficient defensive activity and discouraging redistributive takeovers.

## 6.6 Litigation policy

Litigation is a centralized decision process in which parties expend resources trying to influence courts. Bright-line legal rules and damages limitations reduce discretion and influence costs, though they may also reduce flexibility and individualized justice.

## 6.7 Table 4: Applications of the influence-cost framework

| Application              | Influence-cost logic                                                                  | Design implication                                                                |
|--------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Internal job assignments | Workers care about routes, offices, projects, and roles; discretion creates lobbying. | Use seniority, bidding, rotation, or transparent assignment rules.                |
| Innovation policy        | Incumbents lobby to block innovations with large redistributive effects.              | Limit centralized veto power over experimentation when social downside is small.  |
| Resistance to change     | Employees protect rents created by existing assignments and job characteristics.      | Anticipate redistribution; compensate or redesign rules to reduce resistance.     |
| Vertical integration     | Suppliers spend resources influencing procurement decisions.                          | Integration can reduce influence costs by removing some sourcing discretion.      |
| Takeovers                | Acquirers and target managers spend resources over redistributive control transfers.  | Compensation packages can reduce inefficient defensive or acquisition activity.   |
| Litigation               | Parties spend resources influencing a centralized judge or jury.                      | Bright-line rules and damages limits trade flexibility for lower influence costs. |
| Public bureaucracy       | Public access and broad government discretion invite lobbying.                        | Smaller bureaucracies and limited discretion can reduce rent seeking.             |

# 7 Empirical and theoretical status

## 7.1 Nature of the paper

This is a theory paper, not an empirical paper. It does not estimate influence costs or test its predictions with data. Instead, it provides a conceptual model and a set of applications that

generate testable implications for organizational design.

## **7.2 Potential empirical implications**

The paper suggests several empirical predictions:

- Discretion-limiting rules should be more common when the private stakes for employees are large and firm-value consequences are small.
- Influence activities should increase when internal decisions become more discretionary.
- Standard operating procedures should be especially prevalent for personnel decisions, job assignments, promotions, and other high-redistribution decisions.
- Centralized decision-making should be less attractive when the relevant information benefit is small or when stakeholders can easily lobby the decision maker.
- Change initiatives should face greater resistance when they redistribute working conditions, status, rents, or future career opportunities.

## **7.3 Limitations**

The paper leaves several issues open. It does not empirically measure influence costs. It treats influence activity mainly as costly time allocation, while in real organizations influence can also involve misinformation, coalition formation, social pressure, or strategic delay. It also abstracts from settings where lobbying conveys valuable information to decision makers. Finally, the paper does not fully solve the problem of designing optimal communication institutions, though it points toward that agenda.

## 8 Tables and extracted source blocks

### 8.1 Figure 1: Abstract and Theorem 1 on non-compensating differentials

#### Employment Contracts, Influence Activities, and Efficient Organization Design

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When changing jobs is costly, efficient employment contracts usually fail to compensate workers for the effects of postrhiring events and decisions. Then, when there are executives and managers with authority to make discretionary decisions, affected employees will be led to waste valuable time trying to influence their decisions. Efficient organization design counters this tendency by limiting the discretion of decision makers, especially for those decisions that have large distributional consequences but that are otherwise of little consequence to the organization.

The inference to which we are brought is that the causes of faction cannot be removed, and that relief is only to be sought in the means of controlling its effects. [JAMES MADISON, *The Federalist*]

#### I. Introduction

Experience suggests—and most Western economists believe—that decentralized economic authority such as that found in market economies encourages innovation and promotes efficient resource use. The

I owe a special debt of thanks to John Roberts for allowing me to incorporate the fruits of our joint research into this paper, for suggesting some of the reported applications and references, and for his thorough and constructive commentary on the previous versions. For their helpful comments, I also thank Drew Fudenberg, Bengt Holmstrom, Roger Kormendi, Ed Lazear, Maggie Levenstein, Rick Levin, Meg Myer, Jim Mirrlees, Charlie Plott, Ariel Rubinstein, and Jean Tirole, my research assistants Byung-II Choi and Bernard Desgagne, and two anonymous referees. I gratefully acknowledge the financial support of the National Science Foundation, the John Simon Guggenheim Foundation, the Sloan Foundation, and the University of California, Berkeley.

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(a) Abstract and main claim

Let us consider a family of problems like (CP), parameterized by  $\bar{u}$ . Take  $\hat{u}$  to be any function of  $\bar{u}$  such that  $\hat{u}(\bar{u})$  is always less than  $\bar{u}$ . When does the optimal contract pay full compensating differentials, leaving the employee indifferent among postrhiring events?

**THEOREM 1.** A solution to (CP) exists and makes the employee indifferent among outcomes and just willing to work ( $u_i(w_i^*) = \bar{u}$ ) for every  $\bar{u}$  in the range of  $u_1$  if and only if  $u_1$  is concave and for all  $i$  there exists  $g_i$  such that

$$u_i(w) \equiv u_1(w + g_i) \quad \text{for all } w. \quad (1)$$

*Proof.* It is routine to check that (1) and the concavity of  $u_1$  imply that the optimal contract exists and satisfies  $u_i(w_i^*) = \bar{u}$ ; attention is focused on the reverse implication. Regarding  $w_i^*$  as a function of  $\bar{u}$ , the hypothesis is that  $u_i(w_i^*(\bar{u})) = \bar{u}$  for all  $i$  and all  $\bar{u}$  in the range of  $u_1$ , that is,  $w_i^* = u_i^{-1}$ . The first-order necessary conditions for optimality in (CP) imply that, for all  $i$  and all  $\bar{u}$  in the range of  $u_1$ ,

$$u'_1(w_1^*(\bar{u})) = u'_i(w_i^*(\bar{u})). \quad (2)$$

Then  $w_1^*(\bar{u}) = w_i^*(\bar{u})$ . Hence,  $w_1^*(\bar{u}) \equiv w_i^*(\bar{u}) + g_i$  for all  $\bar{u}$  in the range of  $u_1$ , where the  $g_i$ 's are constants of integration. Then, for any fixed  $w$ ,  $u_1(w + g_i) = u_1[w_i^*(u_i(w)) + g_i] = u_1[w_i^*(u_i(w))] = u_i(w)$ . This holds for all  $w$ , as required.

Given the identity just derived, the second-order necessary conditions imply that  $u''_1(w_1^*(\bar{u})) \leq 0$  for all  $\bar{u}$ , which establishes concavity. Q.E.D.

An optimal contract equates a risk-averse employee's marginal utility of income in the different events  $x$ ; it does not also equate his utilities in the different events unless the employee has ordinal preferences that can be represented by vertically parallel indifference curves in  $(x, w)$ -space. This characterization of ordinal preferences is quite restrictive. When it fails, the optimal contract will not leave the employee indifferent among assignments.

Now let us make an obvious but quite important observation: At the optimal contract, the employee's wages  $w_i^*$  do not depend at all on the gross profit levels  $\pi_i$ . Consequently, *there is no necessary relationship between the employer's ranking of outcomes and the employee's*. Later, when the possibility that the employee can influence the distribution of  $x$  is introduced, this divergence of rankings will become quite important.

(b) Theorem 1 and contract logic

**Read:** The extracted abstract states the central claim that when job changes are costly, employment contracts usually fail to compensate workers for post-hiring events and decisions. Theorem 1 shows why full compensating differentials require special preference conditions.

**Economic message:** The paper first establishes that workers rationally care about assignments, then uses this to explain why discretionary authority invites influence activity.

## 8.2 Figure 2: Examples 1 and 2 on working conditions and human capital

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Theorem 1 is just a starting point. It tells us that full compensating differentials are rarely paid in a large class of contracting models. The remainder of this section is devoted to the development of examples to illustrate the following points: (1) There is not even a general *tendency* for optimal wage schedules to compensate for job characteristics, so that employee job concerns under optimal contracts may be quite pronounced; (2) increases in risk aversion tend to lead to the payment of fuller compensating differentials; (3) employees may care about job attributes under optimal contracts even when, contrary to the simple model just presented, job attributes are not an argument of employee utility functions; and (4) these models lead to plausible predictions about the kinds of preferences among job characteristics that employees may systematically show.

### Example 1: Preference for Good Working Conditions

Let  $x \geq 0$  denote either working conditions or on-the-job consumption<sup>3</sup> and let  $w \geq 0$  denote the wage or at-home consumption. Suppose that the employee's ordinal preferences have the Cobb-Douglas form  $x^\alpha w$  and that his coefficient of relative risk aversion for wage gambles is the constant  $\beta > 0$ , so that the employee is risk averse. These cardinal preferences are represented by  $U(x, w) = \alpha \ln(x) + \ln(w)$  in case  $\beta = 1$  and by  $U(x, w) = (x^\alpha w)^{1-\beta}/(1-\beta)$  in case  $\beta \neq 1$ . Suppose that the employee's initial reservation utility level is  $\bar{u}^{1-\beta}/(1-\beta)$ , with  $\bar{u} > 0$  and  $\hat{u} = -\infty$ . Then the solution to the contracting problem (CP) is  $w(x) = \lambda x^{\alpha(1-\beta)/\beta}$  for some constant  $\lambda$  that depends on the parameters  $\alpha, \beta, \bar{u}$ , and  $(p_1, \dots, p_N)$ . Notice in particular that if  $\beta < 1$ , then  $w(\cdot)$  is actually an increasing function of  $x$ . This establishes that there is no general tendency for optimal contracts to pay even partial compensation for unfavorable working conditions.

In this example, the ordinal utility associated with job  $x$  can be measured by  $x^\alpha w(x) = \lambda x^{\alpha/\beta}$ ; it increases in  $x$  for any level of risk aversion. This last observation is a special case of a general result that has been derived by several authors including Chari (1983), Green and Kahn (1983), and Bergstrom (1986). Their result, applied to this model, holds that if on-the-job consumption is a normal good, then the optimal wage contract will always lead employees to prefer jobs with higher  $x$ .

In example 1, as the coefficient of relative risk aversion  $\beta$  increases, the ordinal utility measure  $x^\alpha w(x) = \lambda x^{\alpha/\beta}$  becomes increasingly flat and converges to the constant  $\bar{u}$ . With increases in risk aversion, the

<sup>3</sup> Stafford and Cohen (1974) supplied one of the earliest economic treatments of on-the-job consumption in a study of how work effort varies during the workday.

(a) Example 1: working conditions

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specific human capital, so the worker's productivity does not depend on whether he remains with his initial employer. There are two possible events in the first period. In the first event, which arises with probability  $r$ , the employer will assign the worker to a task that increases his second-period productivity to an amount  $q > p$ . In the second, which occurs with probability  $1 - r$ , the worker is assigned to a task in which human capital is unchanged, so the worker's second-period marginal product will be  $p$ .

At the beginning of the second period, the worker is free to quit the firm and go to work elsewhere for a wage equal to his current marginal product. This mobility imposes a lower bound on the wage the worker can be paid in the second period. However, there are some market frictions: The employee cannot leave during the first period after learning his job assignment.

Let  $\pi_j$  be an increment to the firm's revenues when event  $j$  occurs and  $w_{ij}$  the corresponding period  $i$  wage. Assume that competition among similar firms drives the expected wage over the two-period contract to be equal to the worker's expected marginal product over that period, which is  $rq + (2 - r)p$ . The model also assumes that the worker can neither borrow nor save (although only the no-borrowing constraint is in fact binding) so that his consumption is equal to his income in each period. Competition among employers will lead them to offer an efficient contract, one that maximizes the worker's utility subject to the maximum expected wage constraint and the constraints on second-period wages:

$$\text{maximize } r[u(w_{11}) + u(w_{21})] + (1 - r)[u(w_{12}) + u(w_{22})] \quad (3)$$

subject to

$$r(w_{11} + w_{21}) + (1 - r)(w_{12} + w_{22}) = rq + (2 - r)p, \quad w_{21} \geq q, w_{22} \geq p,$$

where  $u$  is some strictly concave function.

This is a concave maximization problem with linear constraints, so its optimal solution is fully characterized by a first-order condition. It is not hard to verify that the unique optimal solution has  $w_{21} = q$  and  $w_{11} = w_{12} = w_{22} = p$  with Lagrange multipliers of  $u'(p)$ ,  $r[u'(q) - u'(p)]$ , and zero, respectively, on the three constraints. Thus in each period the employee is paid his current marginal product. An employee who is fortunate enough to be assigned to job 1 acquires valuable human capital but suffers no offsetting wage reduction under the terms of the optimal contract. Consequently, employees prefer job assignments that increase their human capital. The employer's net profit under the contract in event  $j$  is precisely  $\pi_j$ , an amount unrelated to the employee's human capital acquisition. So the employee's interests may conflict with the employer's.

(b) Example 2: human capital

**Read:** Example 1 shows that good working conditions need not be offset by lower pay. Example 2 shows that human-capital-building assignments create benefits for employees even when current wages are not reduced.

**Economic message:** Employees may have strong preferences over assignments for reasons that do not align with firm profits.

### 8.3 Figure 3: Theorem 3 and Theorem 4 on critical jobs and restricted discretion

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The wage policy  $w_i$  that maximizes (4) takes the form  $w_i = w(\Delta_i)$ . Since  $w_i$  does not depend on  $\pi_i$ , there is no necessary relation between the interests of the employer and those of the employee. Thus, as in the previous models, arbitrarily severe conflicts of motives can arise between the employer and the employee. The heuristic optimal wage policy satisfies the rearranged first-order condition

$$\Delta_i = (w_i - b) + (1 - \lambda) \frac{F(w_i - b)}{f(w_i - b)}, \quad (5)$$

where  $\lambda$ , the Lagrange multiplier of the single constraint, is the marginal cost of providing an extra dollar of expected income to the employee. It is clear that  $\lambda$  cannot exceed one. Hence, the right-hand side of (5) is increasing in  $w_i$ , so wages increase with  $\Delta_i$ . Employees prefer to occupy “critical” jobs in which turnover is costly to the employer.

In an appendix of the working paper version of this paper (Milgrom 1987), I present a full formal analysis of this problem without the restriction to simple wage policies used above. In the full model, the employee may report his outside opportunities to his employer, but the truthfulness of any report cannot be assured. The employer can take account of the report in setting wages and termination bonuses and in making layoff decisions; it can also randomize on the basis of the report. The upshot is that none of these additional options is useful to the employer and that the heuristic analysis given above yields the right answer:

**THEOREM 3.** The employer has an optimal policy that requires no randomization or reporting by the employee. The policy establishes a termination bonus  $b$  and, for each assignment  $i$ , a wage  $w_i$ ; the employee quits whenever his outside option pays more than  $w_i - b$ . Under this optimal policy, the wage  $w_i = w(\Delta_i)$  is an increasing function of  $\Delta_i$ .

#### III. When Does It Pay to Restrict Management Discretion?

We now consider a simple model of influence in which the employee allocates his available time  $T$  between two activities, a directly productive activity and an influence activity. If the employee spends time  $t$  at the directly productive activity, then his output will be “high” with probability  $p(t)$  and “low” with probability  $1 - p(t)$ . The organization will earn an extra profit of  $\pi$  if output is high. Assume that  $p'(t)$  is continuous and strictly positive and  $0 < p(t) < 1$  on  $[0, T]$ .

(a) Theorem 3 and influence model setup

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Fix a time allocation  $(s, t)$  and let  $V(s, t)$  be the optimal value of the corresponding “implementation problem”:

$$\max_{\{w_i\}} \sum p_i[u(w_i) - \lambda w_i] + \lambda p(t)\pi \quad (8)$$

subject to  $(s, t)$  solves (6). In standard fashion,  $V(s, t)$  is an upper semicontinuous function on  $[0, T] \times [0, T]$ .

With this notation, the social problem can be expressed as  $\max_{s,t} V(s, t) + \lambda I\gamma(s)$ . The value of this transformed social objective is increasing in  $I$ , so the optimal value is increasing in  $I$  as well.

When there is no decision maker with authority to alter the status quo, the maximal social payoff is

$$\bar{V} = \max_w u(w) - \lambda w + p(T)\pi.$$

**LEMMA.**  $\bar{V} > \max\{V(s, t) | s + t \leq T, s, t \geq 0\}$ .

*Proof.* First, we claim that  $\bar{V} > V(0, T)$ . Indeed,  $\bar{V}$  is the maximal value of the relaxed version of (8) with  $s = 0, t = T$ , and the incentive constraint—that  $(s, t)$  maximizes (6)—omitted. The unique optimum of the relaxed problem has  $u'(w_i) = \lambda$  for all  $i$ . But then  $(0, T)$  does not maximize (6), so the optimal value of the constrained problem is less than the optimal value of the relaxed problem:  $\bar{V} > V(0, T)$ .

Next we claim that  $\bar{V} > V(s, t)$  for all  $(s, t)$  with  $t < T$ . Indeed, the optimal value of the relaxed version of (8) with the incentive constraint omitted is obtained by setting  $u'(w_i) = \lambda$  for all  $i$ , which yields the optimal value  $\bar{V} + \lambda[p(t) - p(T)]\pi$ . Since  $\lambda\pi p' > 0$ , this is less than  $\bar{V}$  for all  $t < T$ , as claimed.

Finally, since  $V$  is upper semicontinuous, there exists a pair  $(s^*, t^*)$  such that

$$V(s^*, t^*) = \max\{V(s, t) | s + t \leq T, s, t \geq 0\}.$$

Whatever that pair is,  $V(s^*, t^*) < \bar{V}$ . Q.E.D.

The optimal value achieved when management has no discretion to authorize a change is  $\bar{V}$ , and  $\max_{s,t} V(s, t) + \lambda I\gamma(s)$  is the optimal value when management does have discretion. In view of the lemma and the boundedness of  $\gamma(\cdot)$ , it is clear that as  $I$  approaches zero it is best to restrict management discretion.

**THEOREM 4.** There exists a pair of parameters  $(I, k)$  such that when these parameters prevail, it is better to eliminate discretion than to provide wage incentives to limit influence activities. Moreover, if  $(I, k)$  is such a pair and if  $I' \leq I$  and  $k' \geq k$ , then  $(I', k')$  is another such pair.

*Proof.* The arguments preceding the theorem establish all its assertions except the assertion that if discretion is optimally permitted for the parameter pair  $(I, k)$  and if  $k' < k$ , then discretion is optimally

(b) Theorem 4: restricting discretion

**Read:** Theorem 3 implies that workers prefer critical jobs because optimal wages rise with the cost of turnover. Theorem 4 states the central organization-design result: discretion should be removed for low-importance, high-redistribution decisions.

**Economic message:** The firm may prefer rules to discretion even though discretion could improve decision quality, because discretion creates influence costs.

## 8.4 Figure 4: Related literature and applications

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on  $\hat{I}(\cdot)$ . Then a result resembling theorem 4 can be obtained in terms of the social importance function  $\hat{I}$  and the real parameter  $k$ .

Finally, several of the assumptions made here have been relaxed by Milgrom and Roberts (1987b), who studied influence activities by workers seeking a desirable job assignment. Their model includes the possibilities of competition among workers, promotions as rewards for past performance, and decision rules that are chosen in advance by management. Despite these differences, their conclusions reinforce the general finding that central decision makers ought not always be allowed full discretion to make optimal decisions given the facts at hand since that leads to excessive influence activity.

### IV. Related Literature

Williamson (1985) and Grossman and Hart (1986) have offered an alternative explanation of the diseconomies of centralized control that emphasizes the hazards that arise from opportunistic behavior by the owner-managers of integrated firms. These theories complement the one presented here, which emphasizes distortions in the behavior of those who inform and advise the executive that accompany increases in executive authority. These two theories are among those integrated into a general transaction costs framework by Milgrom and Roberts (1987a).

The theory here can be viewed as an extension of the rent-seeking theories developed by Tullock (1967), Krueger (1974), Posner (1975), Buchanan (1980), and Bhagwati (1982). These theories hold that government-granted subsidies, tariffs, and monopolies impose welfare losses on society because they lead businesses to waste resources in attempts to win tariff protection or monopoly rights for themselves. The analyses all indicate that government interventions ought sometimes to be limited in order to discourage wasteful rent-seeking activity.

The analysis here extends these theories in two principal respects. First, this is explicitly a cost-benefit analysis. Because those most affected by a decision are often among those best informed about the alternatives and their consequences, or are at least best motivated to discover and analyze the alternatives and their likely consequences, a reasonable theory must allow the possibility that the activities of rent seekers can lead to better decisions. Second, the scope of the theory is expanded beyond the public sector. There are tremendous payoffs in the private sector from "salesmanship"—both the actual commissions earned by salesmen and the gains to having one's ideas accepted or projects adopted or performance evaluated favorably. This analysis substitutes a broad focus on the costs of centralized authority for the usual narrower focus on the costs of government intervention.

(a) Related literature

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creased authority in the merged firm makes their jobs more "critical" in the sense of example 3 of Section II. Mere transfers of the rents earned by the former management to the shareholders and new management do not enhance efficiency, and that part of takeover activity by the acquiring firm and defensive activity by the target firm's management that is simply redistributive is wasteful. Golden parachutes, properly designed, are executive compensation packages that force potential acquirers to reimburse former managers for any lost rents when there is a transfer of control. These discourage inefficient takeovers and reduce both rent seeking by potential acquirers and rent-protecting behavior by existing management. The consequent efficiency gains ultimately benefit the shareholders.

4. *Litigation policy.*—A court trial is a centralized decision process in which the disputants often incur huge costs to effect a redistribution of wealth. As "bright-line" law fades and parties become less sure of the likely outcome of litigation, the discretion of juries and judges correspondingly rises.<sup>6</sup> Damage rules play the role that wages played in this study of influence within firms: Rules limiting damages reduce influence costs at the expense of other objectives such as paying "just" compensation or creating efficient incentives for contractual performance.

### VI. Concluding Remarks

The economic environment described here differs markedly from the neoclassical, perfectly competitive spot contracting environment in which buyers are indifferent at the margin about what they buy, sellers are indifferent about incremental sales, and workers are indifferent about employer decisions. Instead, people care about decisions and attempt to influence them. When decision makers are honest and rational, influence takes the form of suggesting alternatives and supplying information, opinion, and analysis; when they are not, influence may take more insidious forms. Efficient organization design seeks to do what the system of prices and property rights does in the neoclassical conception: to channel the self-interested behavior of individuals away from purely redistributive activities and into well-coordinated, socially productive ones. The success that a society's in-

<sup>6</sup> Even those who are firmly bound to pursue a fixed objective or to adhere to a fixed set of rules have discretion to the extent that they may exercise judgment in interpreting and applying rules, admitting and evaluating evidence, resolving ambiguities, etc. For judges and juries, conflicting precedents and novel circumstances result in increased discretion for decision making, which makes it possible for interested parties to profit from what I have dubbed "influence activities."

(b) Applications and conclusion

**Read:** The related-literature section links the paper to transaction-cost economics and rent-seeking theory. The applications section extends the model to resistance to change, vertical integration, takeovers, golden parachutes, and litigation.

**Economic message:** Influence-cost logic is a general institutional-design principle, not only a theory of internal job assignments.

## 9 Short summary

### 9.1 What the paper does

The paper builds a theory of why employees care about post-hiring decisions and how this creates influence costs inside organizations. It first shows that efficient employment contracts normally do not fully compensate workers for all job characteristics. It then shows that when employees care about decisions, managerial discretion creates incentives for costly influence activities. The organization can use wages to mitigate these costs, but sometimes the better solution is to restrict managerial discretion.

## 9.2 Most important takeaway

The core takeaway is that discretion is not always good. More discretion may improve information use and coordination, but it also raises the value of influencing decision makers. Efficient organizations therefore use rules, seniority systems, procedural limits, and other discretion-reducing institutions when decisions are mainly redistributive rather than value-enhancing.

## 9.3 How to remember the paper

A compact way to remember the paper is:

Discretion is valuable when  $I$  is high, but dangerous when  $k$  is high.

Here  $I$  is the organizational importance of the decision and  $k$  is the redistributive value to affected agents.

# 10 Conclusion

## 10.1 Core conclusion

Milgrom's paper provides a foundational theory of internal rent seeking. It explains why employees lobby, resist change, seek favorable assignments, and care about managerial choices even under efficient contracts. The source is not merely irrational behavior or bad management. It is the interaction of incomplete compensation, costly mobility, and centralized discretionary authority.

## 10.2 Incremental contribution revisited

The paper's major contribution is to connect employment contracting, rent seeking, and organization design. It shows that organizational rules are not merely bureaucratic frictions; they can be efficient devices for suppressing costly influence activities.

## 10.3 Implications for reading later papers

This paper is useful whenever a research question involves internal allocation, authority, managerial discretion, organizational politics, lobbying, rent seeking, resistance to change, or the trade-off between flexibility and rule-based governance. It is especially relevant for papers studying how organizations allocate opportunities, capital, promotions, contracts, tasks, and attention.

## 10.4 Final takeaway

The paper's final message is institutional: good organizations do not simply maximize managerial flexibility. They design decision rights so that individual self-interest is directed toward productive activity rather than toward influencing redistributive decisions.